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# An integrated interval programming and input–output knowledge model for risk and resiliency management

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ABSTRACT

The COVID-19 pandemic has devastated nearly every sector of the global economy. The research seeks to examine the COVID-19 pandemic's effects on economy and its interdependencies, specifically on mining and minerals, energy, and agriculture sectors. We propose a mathematical model by combining interval programming with input–output knowledge modeling. The input–output model illustrates how pandemic-induced perturbations in one sector spread to others. According to how susceptible they are to disturbances coming from the primary sector (e.g., mining and minerals, agriculture, or energy), a ranking of the impacted sectors can serve as an essential input for risk and resiliency management. The study also determines which sectors are anticipated to suffer significant financial losses due to the pandemic, allowing decision-makers to prioritize their investment plans in light of system-level uncertainty. We further develop four distinct scenarios and analyze the influence of the three most influential sectors on others.

## 1. Introduction

Any economy depends heavily on commodities. Manufacturing, agriculture, and services all depend on the commodities in one way or another [1,2]. The price of commodities directly impacts the nation's Gross Domestic Product (GDP) and correlates with spending on social welfare, reducing poverty, and sustainable production [3,4]. The nations that produce commodities benefit from increased global demand; on the other hand, the countries that consume commodities profit from a decrease in global demand due to lower prices. Rising domestic demand for commodities has increased the demand for raw materials in the manufacturing sector in India [5,6]. Similarly, with the modernization of farming, the agriculture sector has higher requirements for farm equipment, fertilizers, and other chemicals, further stimulating the demand for commodities. Furthermore, the service sector, which contributes around 50% of the GDP, is also a large consumer of commodities, including fuel, industrial metals used in computer hardware, telecommunication, etc. [7]. Another apparent demand for commodities comes from government spending that targets augmenting infrastructure in the economy through large-scale investments in roads, ports, airports, etc. The commodities determine India's present and have a significant ramification on future growth prospects, especially when it aims to be a global leader in the manufacturing and service sector. The high dependence on commodities further gives rise to commodity risk, which emanates from unprecedented events [8,9], including political crises [10], disruptions in the supply chain [11,12]), and wars [13]. Such artificial or natural events lead to price shocks resulting in higher volatility, putting much pressure on revenues and margins. As per the data from the International Monetary Fund (IMF) commodity data portal, the annualized price volatility of several commodities ranged in double digits over the past few years. Further, as per the recent UNCTAD report on the state of commodity dependence, India's commodity imports grew from 156.57 billion to 264.459 billion United States Dollars (USD) from 2009 to 2019 [14].

India depends on imports to meet internal and external demand for energy, industrial metals, precious metals, etc., making it more vulnerable to commodity price shocks in the external markets [15]. Policymakers have been grappling with the issue of commodity risk management [16]. The price shocks adversely affect specific sectors and spill over to other related sectors, thereby impacting the economy's overall growth. Additionally, because of their greater connectedness,

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they are more susceptible to disruptive occurrences that are brought on by both natural and human-caused factors [17–19]. In the same context, the study examines the impact of a pandemic on the operability of important sectors by employing Input–Output Modeling and its interdependencies on the Indian economy. Based on the research gaps highlighted in Section 2.2, the study develops a research question (RQ):
RQ1. What will be the effect of a pandemic on agriculture, energy, and mining and minerals sector, and how will their ripple effect spread to the other facets of the Indian economy and have an impact on them.?

Based on the research question, the study formulates the following research objectives (ROs):
RO1. To investigate and assess the sectors' interdependence.
RO2. To assess the pandemic's impact on the Indian economy for matrices like economic loss and interoperability to the Indian economy.
RO3. To identify and rank the Indian economy's most susceptible sectors, with the purpose of resilience management.

The concept of interdependency analysis is used to study the ripple effects caused by many types of disruptions, such as terrorist attacks, mishaps, and natural disasters. To assess economic losses and inoperability imposed by lockdown-induced demand degradations in the agriculture, energy, mining, and minerals sectors on other sectors, this study applies explicitly interdependency analysis. These three sectors have been specifically studied as they are the major suppliers of raw materials to other sectors, and therefore, disruptions in these sectors lead to several risks, including commodity price risk. The study adds to the body of knowledge by studying the impact analysis of demand reductions in agriculture, energy, and mining and minerals sectors over other sectors has been measured with regards to economic loss and inoperability. The simulations of economic loss and inoperability in four scenarios have been estimated. The results can be helpful to policymakers in determining which sectors are more vulnerable to commodity risks arising due to such unforeseen events [20–22]. The variability observed across the sectors can be useful in prioritizing policy interventions to mitigate the impact of future disruptions to address current high-priority national goals of achieving sustainable growth (23; Niti Aayog, 2020; 24).

The remaining parts of the study are arranged as follows. The literature review is shown in the section after this. Section 3 discusses the research methods and the data used in empirical estimation, followed by a discussion of empirical results in Section 4. The discussions and implications of the empirical findings are discussed in Section 5, followed by conclusions, limitations and avenues for future research in Section 6.

## 2. Literature review

The existing section is divided into two different parts. The first section illustrates studies relevant to global commodities, commodity risk management and pandemics. The second section reflects the research gaps that exist in the literature.

### 2.1. Studies specific to commodity risk management and pandemic

In the era of globalization, commodities have evolved as one of the most important asset classes [25]. Commodities are primarily categorized under energy, metals and agricultural products. Business profitability relies heavily on risk management strategies to fence against uncertainty about future cash flows. The pandemic has damaged almost all elements of the global economy, particularly the commodity futures markets, because of disruption to global supply chains. The fluctuation in commodity prices is a potential risk for organizations with global supply chains. Considering the negative consequences of a pandemic on the economies globally, it is anticipated that the flow of information among commodities and uncertainties will be changed. Several studies related to commodity risk management and the pandemic are highlighted below:

Dai et al. [26] performed an analysis to analyze the significance of the pandemic on commodity price jumps in the context of China. The results from the study reflect various factors that impact the sensitivity of price jumps to the pandemic. Similarly, Memon et al. [27] presented a study to analyze the relationship between energy efficiency and herding behavior, agriculture and metal markets adopting multifractal detrended fluctuation method. The results from the study reveal the presence of various degrees of significant multifractality patterns can be seen in the commodity markets. Various metals and energy stocks' safe-haven contributions to global equity markets during the pandemic are highlighted [28]. The analysis reveals that conventional and green energy stocks do not perform as safe-haven against global equity markets. Further, under a risk management paradigm, Foroni et al. [29] examined the functional relationships between energy, agricultural, and metal commodities. A backtesting technique was needed for the study to choose the most suitable model for commodities.

An analysis of the return spillovers between commodities to look into the possibility of diversifying commodities away from financial markets [30]. Also, Cui et al. [31] used wavelet coherence and the quantile connectedness technique to conduct a study to assess the dynamic linkages between China's oil and commodity futures markets. The analysis's findings show that in the context of China, there are erratic and weakly correlated relationships between oil and commodity futures. A study developed an effective price risk management strategy for investment portfolios [32]. The detailed energy price volatility analysis was performed using the Markov-Switching GARCH methodology.

Balster and Friedrich [33] highlighted a calibrated dynamic Multi-scale Multi-Regional Input–Output (MSMRIO) framework in the German food supply chain context. The study discovers the commodity flows and required transport capacity considering disruptions. Similarly, a study employed Extreme Value Theory (EVT) considering the commodity markets [34]. The study concentrated on the prices of Gold and Silver, considered one of the chosen investments over centuries. Further, Chkili et al. [35] highlighted a study based on commodity volatility forecasting and risk management considering asymmetry and long memory. The study adopted linear and nonlinear GARCH-type models to perform the analysis. Table 1 summarizes several studies related to pandemic and commodity risk management.

### 2.2. Research gaps

The existing literature represents the studies related to business risk management, commodity markets, and energy markets in the wake of pre and post-pandemic. Several potential studies on risk management and commodity markets are presented in Table 1. Further, few studies exist in the literature that considered both commodity risk management and the pandemic crisis. As a result, the present study expects to cater to the gap by suggesting an Input–Output Modeling for Commodity Risk Management in the wake of the pandemic.

## 3. Research methodology

This section discusses the data, input–output modeling, and interval programming. The significant benefit of using input–output models' is their internal consistency. Any changes in the final demands can all be tracked easily. The input–output model makes constrictive—assumptions, such as that every company in a given industry uses the same manufacturing equipment and creates the same goods [45]. The ability to analyze induced, indirect, and direct impacts using input–output models makes them essential. The dealership's employees will have less money to spend elsewhere as a result of their reduced pay, which is the secondary effect. The idea of indirect impact focuses on how altering inputs may indirectly alter output. Last but not least, direct impact shows how a change in inputs directly affects a change in outputs. There is no defined rule for deciding how many types

**Table 1**
Recent literature relevant to pandemic and commodity risk management is evident in the literature.

| Objective | Methodology | Outcome | Limitation/Remarks | References |
|---|---|---|---|---|
| To research how the COVID-19 outbreak has affected the world's equity and commodity markets. | Time series regression model | The results from the study reflect that option-based implied volatility is the best standard of market volatility and investors' fear. | The study is limited to experts' recommendations. | [36] |
| The study analyzed the influence of COVID-19 on commodity future volatilities. | ARMA-GARCH models | The results from the study suggest that pandemic event has long-run volatilities for future returns. | The model is limited to a certain duration. | [37] |
| The study evaluated the dynamics of return connectedness among potential commodity assets and financial assets. | Vector Auto Regressive Model (V.A.R.) | The results from the study recommend that investors and policymakers practice flexible regulatory and portfolio allocation strategies. | The study is limited to the US and China | [38] |
| The study suggested a model among global commodities and uncertainties at various frequencies adopting information flow theory. | Rényi effective transfer entropy and CEEMDAN | The results from the study highlight pertinent inferences for policy decision-making and risk management schemes for global commodities and market volatilities. | The study has the limitation of adopting five uncertainty indices. | [39] |
| This study discusses Business Risk Management (BRM) programs in managing risks through the COVID-19 pandemic. | Literature Review | Study suggest that BRM programs do not obstruct the competitive position of Canadian farmers. | The study is limited to the Canadian agriculture sector | [40] |
| The study focused on how the COVID-19 epidemic affected the energy markets. | Mixing-Data Sampling(MIDAS) methods | The results from the study reflect policymakers to analyze the impact of a pandemic on spillover effects in energy markets. | The study is limited to providing implications to energy markets. | [41] |
| The study examined the pandemic-related risk profile of the global commodities network as represented by the markets for commodities with and without energy. | Neural network quantile regression approach | Results reflect that energy commodities contribute most to the systemic risk during the post-COVID period. | The study is limited to energy and nonenergy commodity markets. | [42] |
| This study examines the dynamic network connectivity between commodity futures and stock markets. | BEKK–GARCH model | The study results reflect that the oil market is the most susceptible and affected by the two crises. | The study is limited to five financial markets and two commodities. | [43] |
| The paper analyzes the time-frequency connection between the markets for agricultural commodities and crude oil prices, as well as the spillover effects between the two. | Spillover index and wavelet coherence model | The study's findings show that there is a large amount of variation in the degree of time-dependent spillover to crude oil prices between marketplaces for agricultural commodities. | The study is limited to a certain duration. | [44] |

of economic activities, ultimate demand, and value-added to include or exclude while building the model. Hence, input–output models have various degrees of openness and closure. The model's analytical intent heavily influences this. Its applications have grown beyond the simplest, which describes interindustry interactions, to more complex and specialized evaluations of economic systems. Applying the input–output model includes sensitivity analysis, feasibility study, economic and policy impact analysis, planning scenarios and forecasting [46]. The transactions table must be mathematically manipulated for each of these studies. To quantify multiplier effects within an economic system, input–output models are frequently employed as an alternative to Keynesian models and simple economic bases.

Interval programming and input–output modeling are used in the study to create a mathematical model that can be used to analyze the effects of the pandemic.

### 3.1. Data collection

In this study, the data has been collected for India from Data Library called A.D.B. Input–Output Economic Indicators. To perform the analysis, 35 different sectors were chosen. Sector Code (SC) information is shown in Appendix for the list of sectors. Four sectors were eliminated from the last achievable model because the full data set was not available. The following sectors are: C31, C32, C33, and C35.

### 3.2. input–output modeling (I-O Model)

It is used to analyze the transactions among consumers and producers in the Indian economy. Indian economy consists of $n$ number of interdependent sectors and a linear equation that investigates the links among different economic sectors. Notations used while developing the linear equations for the mathematical model are presented as follows:

$n$: total economic sectors in India
$x_i$ total output of sector $i$
$x_j$: total output of sector $j$
$c_i$: final demand of sector $i$
$x_{ij}$: intermediate consumption (sector $i$ to sector $j$)
$a_{ij}$: sector $i$'s input demand for sector $j$
where $i, j = 1, 2, \ldots, n$.

The proportionality presumption is the basis of Leontief's I-O model formulation, which results in the equation shown below:

$$x_{ij} = a_{ij} x_i \quad \forall i; j: \tag{1}$$

Furthermore, depending on whether it is consumed for final purposes ($c_i$) or for intermediate purposes ($x_{ij}$), a sector's $i$ complete production is consumed.

$$x_i = \sum_j x_{ij} + c_i \quad \forall i: \tag{2}$$

Using Eq. (1) and Eq. (2), the Leontief I-O Model can be achieved. In the following formulation, a certain commodity will be produced by each sector. Products, which can be either goods or services, are considered commodities.

$$x = Ax + c \tag{3}$$

$$x_i = \sum_j a_{ij} x_{ij} + c_i \tag{4}$$

The input–output coefficient matrix can be represented as:

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & & \vdots \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix} \tag{5}$$

The output $x$ vector and transaction matrix $Z$ have been used to determine the technology matrix coefficient.

$$a_{ij} = \frac{Z_{ij}}{x_j} \tag{6}$$

$$L = (I - A)^{-1} \tag{7}$$

Eq. (7) represents the total requirement matrix, the Leontief Inverse matrix. After simplification, Eq. (3) can be represented as

$$\begin{aligned} x &= Ax + c \\ c &= x - -A{:}x \\ c &= x{:}(1 - A) \\ x &= \frac{C}{(1 - A)} \end{aligned} \tag{8}$$

$$x = x{:}(1 - A)^{-1} \tag{9}$$

$$x = Lc \tag{10}$$

### 3.2.1. input–output (I-O) model for inoperability

The system's failure to serve its goal is known as inoperability. There are several fields where Leontief's I-O model is found and applied. It makes use of dimensionless numbers in the range of 0 and 1, where 0 represents the system's normal state, intermediate values, and partial failure, and 1 represents the system's complete collapse. Haimes [47] explains the initial IIM concept, which is used to examine the connections between the infrastructure. IIM was used by Santos [48] to do an interdependency analysis to look at the short- and long-term effects of terrorism on the American economy. Following is the developed IIM structure:

$$q = A^* \times q + c^* \tag{11}$$

Where,

$q$: The sector's unmet production compared to anticipated production is measured by the interoperability vector. Additionally, it is shown as a normalized economic loss.

$A^*$: The sector connections are represented by the interdependency matrix, and

$c*$: Perturbation vector represents final demand.

### 3.2.2. Inoperability

The difference between the ideal production level ($x_i$) and the disturbed production level ($\tilde{x}_i$) is used to calculate the level of inoperation.

Below is the scaler representation of $q$ in this case:

$$q = [\mathrm{diag}(x)]^{-1}[x - \tilde{x}_i] \tag{12}$$

$$\begin{bmatrix} q_1 \\ \vdots \\ q_i \\ \vdots \\ q_n \end{bmatrix} = \begin{bmatrix} \frac{1}{x_1} & 0 & \dots & \dots & 0 \\ 0 & \ddots & \ddots & & \vdots \\ \vdots & \ddots & \frac{1}{x_i} & & \vdots \\ \vdots & & \ddots & \ddots & 0 \\ 0 & \dots & \dots & 0 & \frac{1}{x_n} \end{bmatrix} \begin{bmatrix} x_1 - \tilde{x}_1 \\ \vdots \\ x_i - \tilde{x}_i \\ \vdots \\ x_n - \tilde{x}_n \end{bmatrix} \tag{13}$$

As a result, the $q$ vector's elements are expressed as:

$$q_i = \frac{(x_i - \tilde{x}_i)}{x_i} \tag{14}$$

### 3.2.3. Interdependency matrix

The Leontief technology matrix A is related to this matrix. A pairwise evaluation of each sector is shown here by the matrix $n \times n$. These notations are from the I-O model.

$$A^* = \mathrm{diag}(x)^{-1} : A : \mathrm{diag}(x) \tag{15}$$

$$\begin{bmatrix} \frac{1}{x_1} & 0 & \dots & \dots & 0 \\ 0 & \ddots & \ddots & & \vdots \\ \vdots & \ddots & \frac{1}{x_i} & & \vdots \\ \vdots & & \ddots & \ddots & 0 \\ 0 & \dots & \dots & 0 & \frac{1}{x_n} \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & & \vdots \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 & 0 & \dots & \dots & 0 \\ 0 & \ddots & \ddots & & \vdots \\ \vdots & \ddots & x_i & & \vdots \\ \vdots & & \ddots & \ddots & 0 \\ 0 & \dots & \dots & 0 & x_n \end{bmatrix} \tag{16}$$

The interdependency matrix's coefficient is determined in scalar form using Eq. (16).

$$a^*_{ij} = a_{ij}\left(\frac{x_i}{x_j}\right) \tag{17}$$

Here, $a^*_{ij}$ stands for the greater interoperability that sector $j$ offers to sector $i$. IIM models employ traditional economic I-O data, as shown by Thekdi and Santos [49].

### 3.2.4. Perturbation

The perturbation vector, denoted by the symbol c* in this case, is determined using the following equation:

$$c^* = [\mathrm{diag}(x)]^{-1}[c - \tilde{c}_i] \tag{18}$$

$$\begin{bmatrix} c^*_1 \\ \vdots \\ c^*_i \\ \vdots \\ c^*_n \end{bmatrix} = \begin{bmatrix} \frac{1}{x_1} & 0 & \dots & \dots & 0 \\ 0 & \ddots & \ddots & & \vdots \\ \vdots & \ddots & \frac{1}{x_i} & & \vdots \\ \vdots & & \ddots & \ddots & 0 \\ 0 & \dots & \dots & 0 & \frac{1}{x_n} \end{bmatrix} \begin{bmatrix} c_1 - \tilde{c}_1 \\ \vdots \\ c_i - \tilde{c}_i \\ \vdots \\ c_n - \tilde{c}_n \end{bmatrix} \tag{19}$$

Where, degraded demand $\tilde{c}_i$ and final demand $c$ are denoted. The following equation can be used to obtain the coefficients of the perturbation vector in scaler form:

$$c^*_i = \frac{(c_i - \tilde{c}_i)}{x_i} \tag{20}$$

### 3.2.5. Economic loss

The combination of ($x_i$) the ideal output level and ($q_i$) the inoperability led to this economic loss. This vector's components stand for the monetary cost of the operational inefficiency of sector $i$.

Economic Loss = $x$

$$\text{Economic Loss} = x_i \times q_i \tag{21}$$

Inoperability and economic loss are critical components of managing risks and constructing resilience in complex systems. Inoperability acts as a measure of system vulnerability and is considered an element in risk assessment. Similarly, economic loss is a metric for impact assessment. Incorporating economic loss and inoperability shows that the suggested mathematical model considers both the technical and financial aspects of risk and resilience. This multifaceted approach gives insightful findings that direct practical risk management tactics.

### 3.3. Interval programming (IP)/ range programming

By setting constraints on mathematical operations, interval programming is a mathematical technique for measuring and rounding mistakes ([50]). It provides a straightforward method for expressing the data uncertainty using upper and lower bounds. Interval programming is a mathematical technique that guarantees accurate and dependable output. It provides a variety of probable values for the quantity that is supplied. This paper examines how the epidemic has affected the economy and its interconnections. It integrates IP with IIM in order to deal with these uncertainties. The interval IIM approach is used in the study to measure how the perturbations affect various sectors. Several IP operations are explained below for further information: Kaufman and Gupta [51].

Suppose, $X = x^L; x^U$ and $Y = y^L; y^U$. The following operation holds:

$$X + Y = x^L + y^L \; ; \; x^U + y^U \tag{22}$$

$$X - Y = [\, x^L - y^L \; ; (x^U - y^U)] \tag{23}$$

$$X * Y = [\min \; x^L y^L; x^U y^L; x^L y^U; x^U y^U \; ; \max(x^L y^L; x^U y^L; x^L y^U; x^U y^U)] \tag{24}$$

$$A/B = [\min \; x^L/y^L; x^U/y^L; x^L/y^U; x^U/y^U \; ; \; \max(x^L/y^L; x^U/y^L; x^L/y^U; x^U/y^U)] \tag{25}$$

## 4. Results and analysis

India is one of the fastest developing countries in the world, contributing around seven percent of the global GDP. The pandemic has impacted the economies worldwide, and India is not different from other countries [52]. The percentage demand reduction is considered as an input to the model. In this study, the most promising sectors, mining and minerals, agriculture and energy, are considered to analyze their impact on other sectors. The detailed results are described in the following sub-sections.

### 4.1. Scenario description

The case study in this manuscript is an analysis during the lockdown period due to the first wave of the COVID-19 pandemic. The Indian government announced a lockdown with effect from March 18, 2020, for a period of over one month. Post a month, the government relaxed the restrictions in a phased manner. Thus, we have considered the lockdown period of 30 days from March 18, 2020, to April 18, 2020, for inoperability analysis.

### 4.2. Inoperability and economic loss rankings

Due to demand decreases brought on by disruptive events, this study produces the demand side inoperability statistic, which measures the percentage difference between the sector's business as usual and output level. In this study, the lockdown caused by the pandemic is considered a disruptive event that impacts production levels. This study considers the total lockdown period of 54 ± 15 days as a perturbation. The reason behind the selection of 54 days is the total effective lockdown period announced by the Indian Government in the wake of COVID-19. The phase-wise lockdown period announced by the government during the first wave of COVID-19 is presented in Table 2. This study analyzes the ripple effect of agriculture, energy, and mining and minerals sectors on other sectors. A total of 31 sectors out of 35 sectors have been included and presented in Appendix. The annual data of these sectors, specifically for the year 2020, have been considered. The impact analysis of demand reductions in one sector over other sectors has been calculated for economic loss and inoperability. These result in four different scenarios: Scenario 1 – Impact of demand reduction in the agriculture sector, Scenario 2 – Impact of demand reduction in the energy sector, Scenario 3 – Impact of demand reduction in mining and minerals sector and Section 4 – Impact of demand reduction in all of the three sectors simultaneously. The average inoperability and economic loss for each sector in all scenarios have been presented in Table 3. The 10 most economically impacted sectors in order of severity namely Agriculture, hunting, forestry, and fishing (C1), Electricity, gas, and water supply (C17), Mining and quarrying (C2), Retail trade, except motor vehicles and motorcycles; repair of household goods (C21), Financial intermediation (C28), Chemicals and chemical products (C9), Coke, refined petroleum, and nuclear fuel (C8), Wholesale trade and commission trade, except of motor vehicles and motorcycles (C20), Renting of M and Eq and other business activities (C30) and, Construction (C18), respectively in all of the scenarios is presented in Fig. 1. The average economic losses reported are 148496.56, 16782.68, 45409.67, and 210594.92 (in million $s) in Scenarios 1, 2, 3 and 4, respectively. However, in terms of inoperability, the top ten most affected sectors are presented in Fig. 2. Particularly considering Scenario 4, Electricity, gas, and water supply (C17), Agriculture, hunting, forestry, and fishing (C1), Mining and quarrying (C2), Coke, refined petroleum, and nuclear fuel (C8) and Retail trade, except motor vehicles and motorcycles; repair of household goods (C21) having an inoperability portion of 24.75%, 21.87%, 21.46%, 2.10% & 2.02%, respectively, are the five most vulnerable sectors in terms of inoperability. The results above reflect how economic loss and inoperability in Agriculture, Mining and Minerals, and energy propagate and cause ripple effects on the economic performance of other sectors. Fig. 3 shows the inoperability versus economic loss matrix. The sectors have been divided into three categories: most affected (red), moderately affected (yellow) and least affected (green). Fig. 3 reflects that C1. C2. C17. C8, C21, C9, C28 and C20 are the most affected sectors in terms of inoperability and economic losses.

**Table 2**
Phases and duration of lockdown during the first wave of COVID-19 in India.

| Time period | Days | Phase of lockdown in India |
|---|---|---|
| March 24, 2020–April 14, 2020 | 22 | 1st Phase |
| April 15, 2020–May 3, 2020 | 18 | 2nd Phase |
| May 4, 2020–May 17, 2020 | 14 | 3rd Phase |
| May 18, 2020–May 31, 2020 | 14 | 4th Phase |

## 5. Discussion

In this study, the case of COVID-19 in India has been considered to understand its impact which would aid commodity risk management. The first wave of COVID-19 was disastrous in nature, and everything was locked down. It adversely affected almost every sector which sets the requirement for performing this analysis [53]. It has been estimated that COVID-19 alone caused global economy by $2 trillion

**Table 3**
Average inoperability and economic loss for sectors in 2020.

| Sector code | Economic loss | | | | Inoperability | | | |
|---|---|---|---|---|---|---|---|---|
| | Agri | Mining | Energy | Overall | Agri | Mining | Energy | Overall |
| C1 | 131 545.28 | 70.07 | 102.55 | 131 717.88 | 21.8505% | 0.0116% | 0.0170% | 21.8792% |
| C2 | 364.67 | 115 83.67 | 1009.03 | 129 57.38 | 0.6040% | 19.1873% | 1.6714% | 21.4627% |
| C3 | 1021.91 | 35.05 | 53.38 | 1110.34 | 0.4347% | 0.0149% | 0.0227% | 0.4723% |
| C4 | 324.15 | 33.46 | 57.82 | 415.43 | 0.2311% | 0.0239% | 0.0412% | 0.2962% |
| C5 | 10.77 | 3.36 | 5.13 | 19.26 | 0.0798% | 0.0249% | 0.0380% | 0.1427% |
| C6 | 46.56 | 31.32 | 28.44 | 106.31 | 0.2686% | 0.1807% | 0.1641% | 0.6133% |
| C7 | 111.11 | 27.85 | 85.58 | 224.54 | 0.3655% | 0.0916% | 0.2815% | 0.7387% |
| C8 | 1174.64 | 442.69 | 1563.58 | 3180.91 | 0.7772% | 0.2929% | 1.0346% | 2.1047% |
| C9 | 2547.47 | 413.75 | 278.30 | 3239.52 | 1.5254% | 0.2477% | 0.1666% | 1.9397% |
| C10 | 189.66 | 88.10 | 54.05 | 331.81 | 0.3624% | 0.1684% | 0.1033% | 0.6340% |
| C11 | 87.73 | 69.91 | 70.78 | 228.42 | 0.1359% | 0.1083% | 0.1096% | 0.3538% |
| C12 | 279.45 | 357.81 | 409.61 | 1046.87 | 0.1198% | 0.1534% | 0.1756% | 0.4488% |
| C13 | 112.51 | 271.75 | 179.91 | 564.17 | 0.1542% | 0.3724% | 0.2465% | 0.7731% |
| C14 | 108.73 | 66.89 | 459.17 | 634.79 | 0.1316% | 0.0809% | 0.5556% | 0.7681% |
| C15 | 176.01 | 148.68 | 192.45 | 517.14 | 0.1227% | 0.1036% | 0.1342% | 0.3605% |
| C16 | 47.46 | 25.00 | 61.13 | 133.60 | 0.0868% | 0.0457% | 0.1118% | 0.2443% |
| C17 | 2251.59 | 845.61 | 361 54.00 | 392 51.20 | 1.4202% | 0.5334% | 22.8043% | 24.7578% |
| C18 | 632.53 | 333.68 | 508.89 | 1475.11 | 0.1359% | 0.0717% | 0.1094% | 0.3170% |
| C19 | 168.88 | 17.49 | 59.95 | 246.32 | 1.1150% | 0.1154% | 0.3958% | 1.6263% |
| C20 | 1285.18 | 135.77 | 463.47 | 1884.42 | 1.0922% | 0.1154% | 0.3939% | 1.6014% |
| C21 | 2949.60 | 163.59 | 778.30 | 3891.50 | 1.5351% | 0.0851% | 0.4051% | 2.0253% |
| C22 | 143.69 | 58.03 | 140.71 | 342.44 | 0.2154% | 0.0870% | 0.2110% | 0.5134% |
| C23 | 116.24 | 37.38 | 66.91 | 220.53 | 0.6545% | 0.2105% | 0.3768% | 1.2418% |
| C24 | 0.11 | 24.07 | 2.69 | 26.87 | 0.0003% | 0.0778% | 0.0087% | 0.0868% |
| C25 | 1.04 | 94.92 | 1.17 | 3.16 | 0.0076% | 0.6928% | 0.0085% | 0.0231% |
| C26 | 291.23 | 97.78 | 94.45 | 483.46 | 0.8662% | 0.2908% | 0.2809% | 1.4380% |
| C27 | 205.76 | 95.01 | 388.77 | 689.53 | 0.2552% | 0.1178% | 0.4822% | 0.8552% |
| C28 | 1672.51 | 458.59 | 1586.41 | 3717.51 | 0.8481% | 0.2325% | 0.8045% | 1.8851% |
| C29 | 5.39 | 5.03 | 4.60 | 15.02 | 0.0028% | 0.0026% | 0.0024% | 0.0078% |
| C30 | 543.77 | 574.96 | 448.33 | 1567.05 | 0.1708% | 0.1806% | 0.1408% | 0.4921% |
| C34 | 80.93 | 171.41 | 100.11 | 352.45 | 0.1131% | 0.2396% | 0.1399% | 0.4926% |

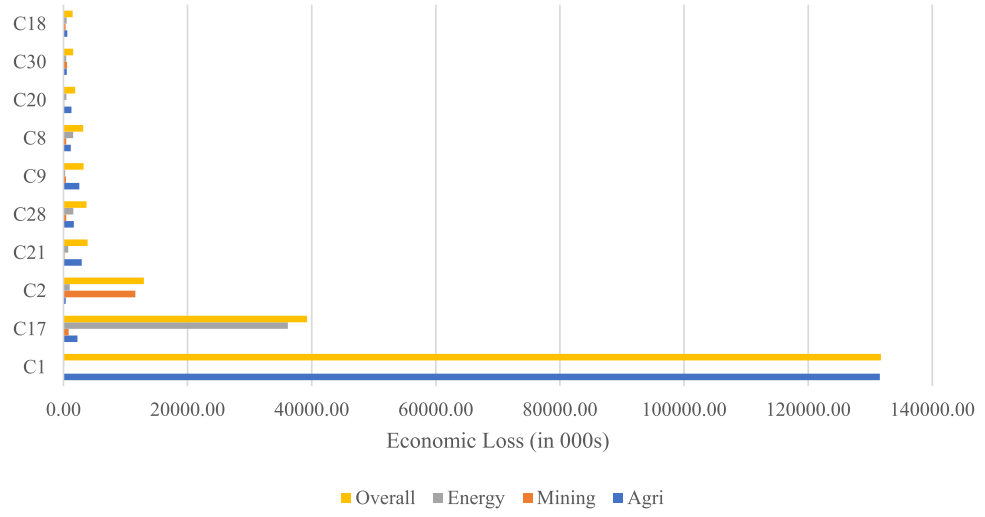


**Fig. 1.** Sector-wise change in Economic Losses of top 10 affected sectors.

in 2020 [54]. The current study analyzed the ripple effect caused by Agriculture, Mining and minerals and, Energy on other sectors through Input–Output Modeling. The results align with the earlier results of Andreoni [55] which advocates the vulnerability of the energy sector due to COVID-19. The agriculture sector should be emphasized due to its dependencies on other sectors. The results have important implications for policymakers in taking initiatives for sectoral recovery. The concerned stakeholders should emphasize pollution prevention as a recovery strategy that would ensure environmental relief and lead to profitability [56]. Today, most countries face sustainable challenges and are committed to addressing the issues [57]. Considering this, several nations have formulated strategies to ensure sustainability, balancing economic, social and environmental aspects. In addition to this, it should be a part of Corporate Social Responsibility (CSR), which would focus on sustainability concerns. The CSR plays a key role in determining the financial performance of the firms [58,59]. The results bring out certain resilient sectors and, therefore, require fewer policy interventions and some sectors are heavily affected and require constant support. Further, identifying the reasons for inoperability and economic losses should be examined in detail to develop appropriate risk management frameworks. The results also point towards developing differential risk management strategies for different sectors. As pointed out by our results C1. C2. C17. C8, C21, C9, C28 and C20 are the most affected sectors in terms of inoperability and economic

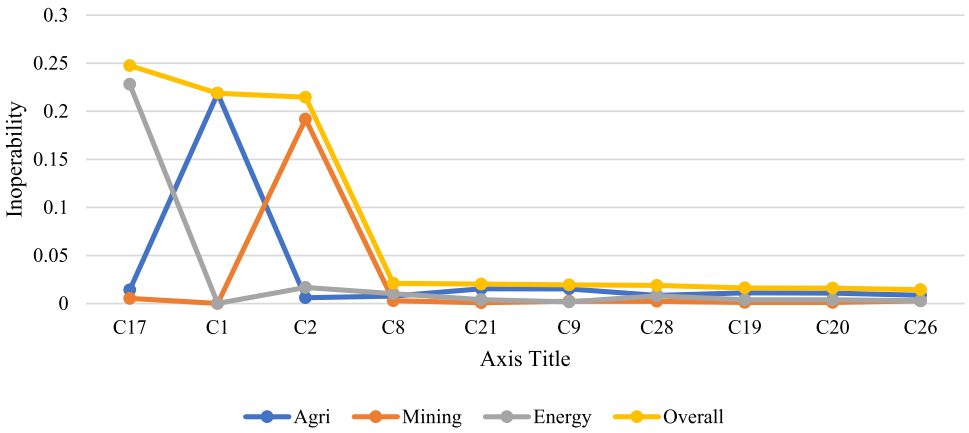


**Fig. 2.** Sector-wise change in Inoperability of top 10 affected sectors.

**Fig. 3.** Inoperability-Economic Loss matrix depicting the severity of effects on sector.. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

losses; therefore, risk management strategies for such sectors should be a priority.

## 6. Managerial implications

The analysis performed in this study has multiple managerial implications. The practitioners can benefit from this study's findings, which aid in better decision-making. The study analyzed the impact of disruptions in the three most promising sectors – Mining and minerals, Agriculture, and Energy – on other sectors in terms of inoperability and economic losses. The relevant parties may also be able to prioritize identifying vulnerable sectors in various scenarios. By developing risk management strategies, the results can be applied to long-term decision-making as well. It would assist decision-makers in addressing COVID-19-like disruptions so that their effect on the economic conditions of other sectors could be minimized. The matrix of Inoperability versus Economic Loss would be useful for locating sectors based on their vulnerability. This can be classified according to high, medium, and low-risk sectors. In the post-COVID-19 era, the practitioners can refer the findings of the current study to ensure the resilience among the system. For an instance, the inoperability-economic loss matrix can be referred to identify the critical sectors which seems to be most affected. The decision-makers may use this information to prepare the sector to fight against potential disaster. Overall, the study offers numerous practical implications that would aid in enhancing our understanding of the ripple effects of one sector on another, prioritizing the vulnerable sectors, developing risk mitigation strategies, and minimizing potential economic losses.

## 7. Conclusions, limitations and avenues for future research

The disruptions cause explicit threats to the economy, especially when it comes to natural disasters such as COVID-19, which is highly uncertain. Such unprecedented events shrink the reaction time, which